# 2.03 Complex Numbers: De Moivre's Theorem - Multiple Angles

Worked Solutions

**1 a)**

$$\begin{aligned}(1+\mathrm{i})^3 &= 1+3\mathrm{i}+3\mathrm{i}^2+\mathrm{i}^3\\ &= 1+3\mathrm{i}-3-\mathrm{i}\\ &= -2+2\mathrm{i}\end{aligned}$$

**1 b)**

$$\begin{aligned}(1+\mathrm{i})^4 &= 1+4\mathrm{i}+6\mathrm{i}^2+4\mathrm{i}^3+\mathrm{i}^4\\ &= 1+4\mathrm{i}-6-4\mathrm{i}+1\\ &= -4\end{aligned}$$

**1 c)**

$$\begin{aligned}(1+\mathrm{i})^5 &= 1+5\mathrm{i}+10\mathrm{i}^2+10\mathrm{i}^3+5\mathrm{i}^4+\mathrm{i}^5\\ &= 1+5\mathrm{i}-10-10\mathrm{i}+5+\mathrm{i}\\ &= -4-4\mathrm{i}\end{aligned}$$

**1 d)**

$$\begin{aligned}(1+\mathrm{i})^6 &= 1+6\mathrm{i}+15\mathrm{i}^2+20\mathrm{i}^3+15\mathrm{i}^4+6\mathrm{i}^5+\mathrm{i}^6\\ &= 1+6\mathrm{i}-15-20\mathrm{i}+15+6\mathrm{i}-1\\ &= -8\mathrm{i}\end{aligned}$$

**2 a)**

$$\begin{aligned}(a+b\mathrm{i})^3 &= a^3+3a^2b\mathrm{i}+3ab^2\mathrm{i}^2+b^3\mathrm{i}^3\\ &= a^3+3a^2b\mathrm{i}-3ab^2-b^3\mathrm{i}\\ &= (a^3-3ab^2)+(3a^2b-b^3)\mathrm{i}\end{aligned}$$

**2 b)**

$$\begin{aligned}(a+b\mathrm{i})^4 &= a^4+4a^3b\mathrm{i}+6a^2b^2\mathrm{i}^2+4ab^3\mathrm{i}^3+b^4\mathrm{i}^4\\ &= a^4+4a^3b\mathrm{i}-6a^2b^2-4ab^3\mathrm{i}+b^4\\ &= (a^4-6a^2b^2+b^4)+(4a^3b-4ab^3)\mathrm{i}\end{aligned}$$

**2 c)**

$$\begin{aligned}(a+b\mathrm{i})^5 &= a^5 + 5a^4b\mathrm{i} + 10a^3b^2\mathrm{i}^2 + 10a^2b^3\mathrm{i}^3 + 5ab^4\mathrm{i}^4 + b^5\mathrm{i}^5 \\ &= a^5 + 5a^4b\mathrm{i} - 10a^3b^2 - 10a^2b^3\mathrm{i} + 5ab^4 + b^5\mathrm{i} \\ &= (a^5 - 10a^3b^2 + 5ab^4) + (5a^4b - 10a^2b^3 + b^5)\mathrm{i}\end{aligned}$$

**3 a)**

$$(\cos\theta + \mathrm{i}\sin\theta)^2 = \cos^2\theta + 2\mathrm{i}\cos\theta\sin\theta - \sin^2\theta$$

**3 b)**

Since $\cos 2\theta + \mathrm{i}\sin 2\theta = (\cos\theta + \mathrm{i}\sin\theta)^2$ Equate real parts:

$$\cos 2\theta = \cos^2\theta - \sin^2\theta \quad \text{as req.}$$

Equate imaginary parts:

$$\sin 2\theta = 2\cos\theta\sin\theta \quad \text{as req.}$$

**4 a)**

$$(\cos\theta + \mathrm{i}\sin\theta)^3 = \cos^3\theta + 3\mathrm{i}\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - \mathrm{i}\sin^3\theta$$

**4 b)**

Since $\cos 3\theta + \mathrm{i}\sin 3\theta = (\cos\theta + \mathrm{i}\sin\theta)^3$ Equate real parts:

$$\begin{aligned}\cos 3\theta &= \cos^3\theta - 3\cos\theta\sin^2\theta \\ &= \cos^3\theta - 3\cos\theta(1 - \cos^2\theta) \\ &= \cos^3\theta - 3\cos\theta + 3\cos^3\theta \\ &= 4\cos^3\theta - 3\cos\theta \quad \text{as req.}\end{aligned}$$

Equate imaginary parts:

$$\begin{aligned}\sin 3\theta &= 3\cos^2\theta\sin\theta - \sin^3\theta \\ &= 3(1 - \sin^2\theta)\sin\theta - \sin^3\theta \\ &= 3\sin\theta - 3\sin^3\theta - \sin^3\theta \\ &= 3\sin\theta - 4\sin^3\theta \quad \text{as req.}\end{aligned}$$

**4 c)**

$$\begin{aligned}\tan 3\theta &= \frac{\sin 3\theta}{\cos 3\theta} \\ &= \frac{3\sin\theta - 4\sin^3\theta}{4\cos^3\theta - 3\cos\theta}\end{aligned}$$

Divide numerator and denominator by $\cos^3\theta$:

$$= \frac{3\tan\theta\sec^2\theta - 4\tan^3\theta}{4 - 3\sec^2\theta}$$

Use the identity $1 + \tan^2\theta = \sec^2\theta$:

$$= \frac{3\tan\theta(1 + \tan^2\theta) - 4\tan^3\theta}{4 - 3(1 + \tan^2\theta)}$$
$$= \frac{3\tan\theta + 3\tan^3\theta - 4\tan^3\theta}{4 - 3 - 3\tan^2\theta}$$
$$= \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$$

**5 a)**

Let $f(x) = 4x^3 - 3x - 1$

$$f(1) = 4 - 3 - 1 \quad \therefore (x-1) \text{ is a factor}$$
$$4x^3 - 3x - 1 = (x-1)(4x^2 + 4x + 1) = (x-1)(2x+1)^2$$

So $x = 1$ or $x = -\frac{1}{2}$

**5 b)**

$$\cos 3\theta = 1 \Rightarrow 3\theta = 0 \text{ or } 3\theta = 2\pi \text{ so } \theta = 0 \text{ or } \theta = \frac{2\pi}{3}$$

**5 c)**

If $x = \cos\theta$ then $x = \cos 0$ or $x = \cos\frac{2\pi}{3}$ So $x = 1$ or $x = -\frac{1}{2}$

**6 a)**

$$(\cos\theta + \mathrm{i}\sin\theta)^4 = \cos^4\theta + 4\mathrm{i}\cos^3\theta\sin\theta - 6\cos^2\theta\sin^2\theta - 4\mathrm{i}\cos\theta\sin^3\theta + \sin^4\theta$$

Since $\cos 4\theta + \mathrm{i}\sin 4\theta = (\cos\theta + \mathrm{i}\sin\theta)^4$ Equate real parts:

$$\begin{aligned}\cos 4\theta &= \cos^4\theta - 6\cos^2\theta\sin^2\theta + \sin^4\theta \\ &= \cos^4\theta - 6\cos^2\theta(1 - \cos^2\theta) + (1 - \cos^2\theta)(1 - \cos^2\theta) \\ &= \cos^4\theta - 6\cos^2\theta + 6\cos^4\theta + 1 - 2\cos^2\theta + \cos^4\theta \\ &= 8\cos^4\theta - 8\cos^2\theta + 1 \quad \text{as req.}\end{aligned}$$

**6 b)**

$$x^4 - x^2 = -\frac{1}{4} \Rightarrow 8x^4 - 8x^2 = -2 \Rightarrow 8x^4 - 8x^2 + 1 = -1$$

Since $\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 2$ then let $x = \cos\theta$

$$\cos 4\theta = -1 \text{ has solutions } \theta \in \{\ldots, \tfrac{\pi}{4}, \tfrac{3\pi}{4}, \tfrac{5\pi}{4}, \tfrac{7\pi}{4}, \ldots\}$$

Choose distinct values:

$$x = \cos\frac{\pi}{4} = \frac{\sqrt{2}}{2} \text{ or } x = \cos\frac{3\pi}{4} = -\frac{\sqrt{2}}{2}$$
$$\therefore x = \frac{\sqrt{2}}{2} \text{ or } x = -\frac{\sqrt{2}}{2}$$

**7 a)**

$$(\cos\theta + \mathrm{i}\sin\theta)^4 = \cos^4\theta + 4\mathrm{i}\cos^3\theta\sin\theta - 6\cos^2\theta\sin^2\theta - 4\mathrm{i}\cos\theta\sin^3\theta + \sin^4\theta$$

Since $\cos 4\theta + \mathrm{i}\sin 4\theta = (\cos\theta + \mathrm{i}\sin\theta)^4$ Equate imaginary parts:

$$\sin 4\theta = 4\cos^3\theta\sin\theta - 4\cos\theta\sin^3\theta \quad \text{as req.}$$

**7 b)**

$$\begin{aligned}
\int_0^{\frac{3\pi}{4}} (\cos^3\theta\sin\theta - \cos\theta\sin^3\theta)\,\mathrm{d}\theta &= \int_0^{\frac{3\pi}{4}} \frac{1}{4}\sin 4\theta\,\mathrm{d}\theta \\
&= \left[-\frac{1}{16}\cos 4\theta\right]_0^{\frac{3\pi}{4}} \\
&= -\frac{1}{16}\cos\left(4\times\frac{3\pi}{4}\right) - \left(-\frac{1}{16}\cos(4\times 0)\right) \\
&= \frac{1}{16} - -\frac{1}{16} = \frac{1}{8}
\end{aligned}$$

as req.

**8)**

$$(\cos\theta + \mathrm{i}\sin\theta)^3 = \cos^3\theta + 3\mathrm{i}\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - \mathrm{i}\sin^3\theta$$

Since $\cos 3\theta + \mathrm{i}\sin 3\theta = (\cos\theta + \mathrm{i}\sin\theta)^3$ Equate real parts:

$$\begin{aligned}
\cos 3\theta &= \cos^3\theta - 3\cos\theta\sin^2\theta \\
&= \cos^3\theta - 3\cos\theta(1 - \cos^2\theta) \\
&= \cos^3\theta - 3\cos\theta + 3\cos^3\theta \\
&= 4\cos^3\theta - 3\cos\theta
\end{aligned}$$

Equate imaginary parts:

$$\begin{aligned}
\sin 3\theta &= 3\cos^2\theta\sin\theta - \sin^3\theta \\
&= 3(1 - \sin^2\theta)\sin\theta - \sin^3\theta \\
&= 3\sin\theta - 3\sin^3\theta - \sin^3\theta \\
&= 3\sin\theta - 4\sin^3\theta
\end{aligned}$$

So that

$$\begin{aligned}
\cot 3\theta &= \frac{\cos 3\theta}{\sin 3\theta} \\
&= \frac{4\cos^3\theta - 3\cos\theta}{3\sin\theta - 4\sin^3\theta}
\end{aligned}$$

Divide numerator and denominator by $\sin^3\theta$:

$$= \frac{4\cot^3\theta - 3\cot\theta\,\mathrm{cosec}^2\theta}{3\,\mathrm{cosec}^2\theta - 4}$$

Use the identity $1 + \cot^2\theta = \sec^2\theta$:

$$\begin{aligned} &= \frac{4\cot^3\theta - 3\cot\theta(1+\cot^2\theta)}{3(1+\cot^2\theta) - 4} \\ &= \frac{4\cot^3\theta - 3\cot\theta - 3\cot^3\theta}{3 + 3\cot^2\theta - 4} \\ &= \frac{\cot^3\theta - 3\cot\theta}{3\cot^2\theta - 1} \quad \text{as req.} \end{aligned}$$

**9 a)**

$$(\cos\theta + \mathrm{i}\sin\theta)^5 = \cos^5\theta + 5\mathrm{i}\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta - 10\mathrm{i}\cos^2\theta\sin^3\theta + 5\cos\theta\sin^4\theta + \mathrm{i}\sin^5\theta$$

Since $\cos 5\theta + \mathrm{i}\sin 5\theta = (\cos\theta + \mathrm{i}\sin\theta)^5$ Equate real parts:

$$\begin{aligned} \cos 5\theta &= \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta \\ &= \cos^5\theta - 10\cos^3\theta(1-\cos^2\theta) + 5\cos\theta(1-\cos^2\theta)(1-\cos^2\theta) \\ &= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta(1 - 2\cos^2\theta + \cos^4\theta) \\ &= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta - 10\cos^3\theta + 5\cos^5\theta \\ &= 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta \quad \text{as req.} \end{aligned}$$

**9 b)**

Equate imaginary parts:

$$\begin{aligned} \sin 5\theta &= 5\cos^4\theta\sin\theta - 10\cos^2\theta\sin^3\theta + \mathrm{i}\sin^5\theta \\ &= 5(1-\sin^2\theta)(1-\sin^2\theta)\sin\theta - 10(1-\sin^2\theta)\sin^3\theta + \sin^5\theta \\ &= 5\sin\theta(1 - 2\sin^2\theta + \sin^4\theta) - 10\sin^3\theta + 10\sin^5\theta + \sin^5\theta \\ &= 5\sin\theta - 10\sin^3\theta + 5\sin^5\theta - 10\sin^3\theta + 10\sin^5\theta + \sin^5\theta \\ &= 16\sin^5\theta - 20\sin^3\theta + 5\sin\theta \quad \text{as req.} \end{aligned}$$

**9 c)**

$$(1+\tan^2\theta)(1+\tan^2\theta) = 1 + 2\tan^2\theta + \tan^4\theta$$

**9 d)**

$$\begin{aligned} \tan 5\theta &= \frac{\sin 5\theta}{\cos 5\theta} \\ &= \frac{16\sin^5\theta - 20\sin^3\theta + 5\sin\theta}{16\cos^5\theta - 20\cos^3\theta + 5\cos\theta} \end{aligned}$$

Divide numerator and denominator by $\cos^5\theta$:

$$= \frac{16\tan^5\theta - 20\tan^3\theta\sec^2\theta + 5\tan\theta\sec^4\theta}{16 - 20\sec^2\theta + 5\sec^4\theta}$$

$$
\begin{aligned}
&= \frac{16\tan^5\theta - 20\tan^3\theta(1+\tan^2\theta) + 5\tan\theta(1+\tan^2\theta)(1+\tan^2\theta)}{16 - 20(1+\tan^2\theta) + 5(1+\tan^2\theta)(1+\tan^2\theta)} \\
&= \frac{16\tan^5\theta - 20\tan^3\theta - 20\tan^5\theta + 5\tan\theta(1+2\tan^2\theta+\tan^4\theta)}{16 - 20 - 20\tan^2\theta + 5(1+2\tan^2\theta+\tan^4\theta)} \\
&= \frac{16\tan^5\theta - 20\tan^3\theta - 20\tan^5\theta + 5\tan\theta + 10\tan^3\theta + 5\tan^5\theta}{16 - 20 - 20\tan^2\theta + 5 + 10\tan^2\theta + 5\tan^4\theta} \\
&= \frac{\tan^5\theta - 10\tan^3\theta + 5\tan\theta}{1 - 10\tan^2\theta + 5\tan^4\theta} \quad \text{as req.}
\end{aligned}
$$

**9 e)**

$$
\begin{aligned}
1 - 10x^2 + 5x^4 &= x^5 - 10x^3 + 5x \\
\Rightarrow \frac{x^5 - 10x^3 + 5x}{1 - 10x^2 + 5x^4} &= 1
\end{aligned}
$$

Let $x = \tan\theta$ so that $\tan(5\theta) = 1$ $\tan 5\theta = 1$ has solutions $\theta = \frac{\pi}{20} + \frac{k\pi}{5}$ for integer $k$ Choose distinct values:

$$
x = \tan\frac{\pi}{20} \text{ or } x = \tan\frac{\pi}{4} \text{ or } x = \tan\frac{9\pi}{20} \text{ or } x = \tan\frac{13\pi}{20} \text{ or } x = \tan\frac{17\pi}{20}
$$

$$
\therefore x = 0.158 \text{ or } x = 1 \text{ or } x = 6.314 \text{ or } x = -1.963 \text{ or } x = -0.510
$$

**10 a)**

$$
(1 - \cos^2\theta)^3 = 1 - 3\cos^2\theta + 3\cos^4\theta - \cos^6\theta
$$

**10 b)**

$$
\begin{aligned}
&(\cos\theta + \mathrm{i}\sin\theta)^6 \\
&= \cos^6\theta + 6\mathrm{i}\cos^5\theta\sin\theta - 15\cos^4\theta\sin^2\theta - 20\mathrm{i}\cos^3\theta\sin^3\theta + 15\cos^2\theta\sin^4\theta + 6\mathrm{i}\cos\theta\sin^5\theta - \sin^6\theta
\end{aligned}
$$

Since $\cos 6\theta + \mathrm{i}\sin 6\theta = (\cos\theta + \mathrm{i}\sin\theta)^6$ Equate real parts:

$$
\begin{aligned}
&\cos 6\theta \\
&= \cos^6\theta - 15\cos^4\theta\sin^2\theta + 15\cos^2\theta\sin^4\theta - \sin^6\theta \\
&= \cos^6\theta - 15\cos^4\theta(1-\cos^2\theta) + 15\cos^2\theta(1-\cos^2\theta)^2 - (1-\cos^2\theta)^3 \\
&= \cos^6\theta - 15\cos^4\theta + 15\cos^6\theta + 15\cos^2\theta(1 - 2\cos^2\theta + \cos^4\theta) - (1 - 3\cos^2\theta + 3\cos^4\theta - \cos^6\theta) \\
&= \cos^6\theta - 15\cos^4\theta + 15\cos^6\theta + 15\cos^2\theta - 30\cos^4\theta + 15\cos^6\theta - 1 + 3\cos^2\theta - 3\cos^4\theta + \cos^6\theta \\
&= 32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1 \quad \text{as req.}
\end{aligned}
$$

**10 c)**

$$
\begin{aligned}
64x^6 - 96x^4 + 36x^2 - 3 &= 0 \\
\Rightarrow 64x^6 - 96x^4 + 36x^2 - 2 &= 1 \\
\Rightarrow 32x^6 - 48x^4 + 18x^2 - 1 &= \frac{1}{2}
\end{aligned}
$$

Let $x = \cos\theta$ so that $\cos(6\theta) = \frac{1}{2}\ \cos 6\theta = \frac{1}{2}$ has solutions $\theta = \frac{\pi}{18} + \frac{k\pi}{3}$ for integer $k$ and $\theta = -\frac{\pi}{18} + \frac{k\pi}{3}$ for integer $k$ Choose distinct values:

$$x = \cos\frac{\pi}{18} \text{ or } x = \cos\frac{7\pi}{18} \text{ or } x = \cos\frac{13\pi}{18} \text{ or } x = \cos\frac{5\pi}{18} \text{ or } x = \cos\frac{11\pi}{18} \text{ or } x = \cos\frac{17\pi}{18}$$

$$\therefore x = 0.9848 \text{ or } x = 0.3420 \text{ or } x = -0.6428 \text{ or } x = 0.6428 \text{ or } x = -0.3420$$

or $x = -0.9848$ (4 DP)

**11)**

$(\cos\theta + \mathrm{i}\sin\theta)^6$

$= \cos^6\theta + 6\mathrm{i}\cos^5\theta\sin\theta - 15\cos^4\theta\sin^2\theta - 20\mathrm{i}\cos^3\theta\sin^3\theta + 15\cos^2\theta\sin^4\theta + 6\mathrm{i}\cos\theta\sin^5\theta - \sin^6\theta$

Since $\cos 6\theta + \mathrm{i}\sin 6\theta = (\cos\theta + \mathrm{i}\sin\theta)^6$ Equate imaginary parts:

$$\sin 6\theta = 6\cos^5\theta\sin\theta - 20\cos^3\theta\sin^3\theta + 6\cos\theta\sin^5\theta \quad \text{as req.}$$

**12 a)**

$$(\cos\theta + \mathrm{i}\sin\theta)^4 = \cos^4\theta + 4\mathrm{i}\cos^3\theta\sin\theta - 6\cos^2\theta\sin^2\theta - 4\mathrm{i}\cos\theta\sin^3\theta + \sin^4\theta$$

Since $\cos 4\theta + \mathrm{i}\sin 4\theta = (\cos\theta + \mathrm{i}\sin\theta)^4$ Equate imaginary parts: $\sin 4\theta = 4\cos^3\theta\sin\theta - 4\cos\theta\sin^3\theta$ So that

$$\frac{\sin 4\theta}{\sin\theta} = \frac{4\cos^3\theta\sin\theta - 4\cos\theta\sin^3\theta}{\sin\theta}$$
$$= 4\cos^3\theta - 4\cos\theta\sin^2\theta$$

Use the identity $\sin^2\theta + \cos^2\theta = 1$:

$$= 4\cos^3\theta - 4\cos\theta(1 - \cos^2\theta)$$
$$= 4\cos^3\theta + 4\cos^3\theta - 4\cos\theta$$
$$= 8\cos^3\theta - 4\cos\theta$$

**12 b)**

$$\sin 4\theta = \sin\theta \Rightarrow \frac{\sin 4\theta}{\sin\theta} = 1 \Rightarrow 8\cos^3\theta - 4\cos\theta = 1$$
$$8\cos^3\theta - 4\cos\theta - 1 = 0$$

Let $f(x) = 8x^3 - 4x - 1$

$$f\left(-\frac{1}{2}\right) = -1 + 2 - 1 = 0 \quad \therefore (2x+1) \text{ is a factor.}$$

$$8x^3 - 4x - 1 = (2x+1)(4x^2 - 2x - 1)$$

So $8x^3 - 4x - 1 = 0$ has solutions of $x = -\frac{1}{2}$, $x = \frac{1+\sqrt{5}}{4}$ or $x = \frac{1-\sqrt{5}}{4}$ Using $x = \cos\theta$ then

$$\cos\theta = -\frac{1}{2}, \quad \cos\theta = \frac{1+\sqrt{5}}{4} \text{ or } \cos\theta = \frac{1-\sqrt{5}}{4}$$

$$\therefore \theta = \frac{2\pi}{3}, \quad \theta = \frac{\pi}{5} \text{ or } \theta = \frac{3\pi}{5}$$

$$\therefore \theta = 2.09, \quad \theta = 0.628 \text{ or } \theta = 1.88 \quad (3 \text{ SF})$$

**13 a)**

$$(\cos\theta + \mathrm{i}\sin\theta)^5 = \cos^5\theta + 5\mathrm{i}\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta - 10\mathrm{i}\cos^2\theta\sin^3\theta + 5\cos\theta\sin^4\theta + \mathrm{i}\sin^5\theta$$

Since $\cos 5\theta + \mathrm{i}\sin 5\theta = (\cos\theta + \mathrm{i}\sin\theta)^5$ Equate real parts:

$$\begin{aligned}\cos 5\theta &= \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta \\ &= \cos^5\theta - 10\cos^3\theta(1-\cos^2\theta) + 5\cos\theta(1-\cos^2\theta)(1-\cos^2\theta) \\ &= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta(1-2\cos^2\theta+\cos^4\theta) \\ &= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta - 10\cos^3\theta + 5\cos^5\theta \\ &= 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta\end{aligned}$$

So that

$$\begin{aligned}\frac{\cos 5\theta}{\cos\theta} &= \frac{\cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta}{\cos\theta} \\ &= \cos^4\theta - 10\cos^2\theta\sin^2\theta + 5\sin^4\theta\end{aligned}$$

Use the identity $\sin^2\theta + \cos^2\theta = 1$:

$$\begin{aligned}&= \cos^4\theta - 10\cos^2\theta(1-\cos^2\theta) + 5(1-\cos^2\theta)(1-\cos^2\theta) \\ &= \cos^4\theta - 10\cos^2\theta + 10\cos^4\theta + 5 - 10\cos^2\theta + 5\cos^4\theta \\ &= 16\cos^4\theta - 20\cos^2\theta + 5\end{aligned}$$

**13 b)**

$$\cos 5\theta = -\cos\theta \Rightarrow \frac{\cos 5\theta}{\cos\theta} = -1 \Rightarrow 16\cos^4\theta - 20\cos^2\theta + 5 = -1$$

$$16\cos^4\theta - 20\cos^2\theta + 6 = 0$$

Let $f(x) = 16x^4 - 20x^2 + 6$

$$16x^4 - 20x^2 + 6 = 2(2x^2-1)(4x^2-3)$$

So $16x^4 - 20x^2 + 6 = 0$ has solutions of $x = -\frac{1}{\sqrt{2}}$, $x = \frac{1}{\sqrt{2}}$, $x = -\frac{\sqrt{3}}{2}$ or $x = \frac{\sqrt{3}}{2}$ Using $x = \cos\theta$ then

$$\cos\theta = -\frac{1}{\sqrt{2}}, \quad \cos\theta = \frac{1}{\sqrt{2}}, \quad \cos\theta = -\frac{\sqrt{3}}{2} \text{ or } \cos\theta = \frac{\sqrt{3}}{2}$$

$$\therefore \theta = \frac{3\pi}{4}, \quad \theta = \frac{\pi}{4}, \quad \theta = \frac{5\pi}{6} \text{ or } \theta = \frac{\pi}{6}$$

**14 a)**

$$z^3 = (1+3\mathrm{i})^3 = 1 + 9\mathrm{i} - 27 - 27\mathrm{i} = -26 - 18\mathrm{i}$$

**14 b)**

$$\arg z = \arctan\left(\frac{3}{1}\right) = \arctan 3$$

**14 c)**

$$(\cos\theta + \mathrm{i}\sin\theta)^3 = \cos^3\theta + 3\mathrm{i}\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - \mathrm{i}\sin^3\theta$$

Since $\cos 3\theta + \mathrm{i}\sin 3\theta = (\cos\theta + \mathrm{i}\sin\theta)^3$ Equate real parts:

$$\begin{aligned}\cos 3\theta &= \cos^3\theta - 3\cos\theta\sin^2\theta \\ &= \cos^3\theta - 3\cos\theta(1-\cos^2\theta) \\ &= \cos^3\theta - 3\cos\theta + 3\cos^3\theta \\ &= 4\cos^3\theta - 3\cos\theta \quad \text{as req.}\end{aligned}$$

Equate imaginary parts:

$$\begin{aligned}\sin 3\theta &= 3\cos^2\theta\sin\theta - \sin^3\theta \\ &= 3(1-\sin^2\theta)\sin\theta - \sin^3\theta \\ &= 3\sin\theta - 3\sin^3\theta - \sin^3\theta \\ &= 3\sin\theta - 4\sin^3\theta \quad \text{as req.}\end{aligned}$$

**14 d)**

$$\begin{aligned}\tan 3\theta &= \frac{\sin 3\theta}{\cos 3\theta} \\ &= \frac{3\sin\theta - 4\sin^3\theta}{4\cos^3\theta - 3\cos\theta}\end{aligned}$$

Divide numerator and denominator by $\cos^3\theta$:

$$= \frac{3\tan\theta\sec^2\theta - 4\tan^3\theta}{4 - 3\sec^2\theta}$$

Use the identity $1 + \tan^2\theta = \sec^2\theta$:

$$\begin{aligned}&= \frac{3\tan\theta(1+\tan^2\theta) - 4\tan^3\theta}{4 - 3(1+\tan^2\theta)} \\ &= \frac{3\tan\theta + 3\tan^3\theta - 4\tan^3\theta}{4 - 3 - 3\tan^2\theta} \\ &= \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta} \quad \text{as req.}\end{aligned}$$

**14 e)**

From $z = 1 + 3\mathrm{i}$ we have $\arg z = \theta = \arctan 3$ So that $\tan\theta = 3$ Using the identity, this gives

$$\tan 3\theta = \frac{3\times 3 - 3^3}{1 - 3\times 3^2} = -\frac{18}{-26} = \frac{9}{13}$$

Also $z^3 = (1+3\mathrm{i})^3 = -26 - 18\mathrm{i}$ But since we know that for complex number $a + b\mathrm{i}$, its argument $\phi$ satisfies $\tan\phi = \frac{b}{a}$ Using the argument, this gives

$$\tan 3\theta = \frac{\mathrm{Im}(z^3)}{\mathrm{Re}(z^3)} = \frac{-18}{-26} = \frac{9}{13}$$

So the results are consistent.

**15 a)**

$$(\cosh x + \sinh x)^3 = \cosh^3 x + 3\cosh^2 x \sinh x + 3\cosh x \sinh^2 x + \sinh^3 x$$

**15 b)**

$$\begin{aligned}
(\cosh x + \sinh x)^3 &= \left(\frac{\mathrm{e}^x + \mathrm{e}^{-x}}{2} + \frac{\mathrm{e}^x - \mathrm{e}^{-x}}{2}\right)^3 \\
&= \left(\frac{\mathrm{e}^x}{2} + \frac{\mathrm{e}^{-x}}{2} + \frac{\mathrm{e}^x}{2} - \frac{\mathrm{e}^{-x}}{2}\right)^3 \\
&= (\mathrm{e}^x)^3 = \mathrm{e}^{3x} \\
&= \frac{\mathrm{e}^{3x}}{2} + \frac{\mathrm{e}^{-3x}}{2} + \frac{\mathrm{e}^{3x}}{2} - \frac{\mathrm{e}^{-3x}}{2} \\
&= \frac{\mathrm{e}^{3x} + \mathrm{e}^{-3x}}{2} + \frac{\mathrm{e}^{3x} - \mathrm{e}^{-3x}}{2} \\
&= \cosh 3x + \sinh 3x \\
(\cosh x - \sinh x)^3 &= \left(\frac{\mathrm{e}^x + \mathrm{e}^{-x}}{2} - \frac{\mathrm{e}^x - \mathrm{e}^{-x}}{2}\right)^3 \\
&= \left(\frac{\mathrm{e}^x}{2} + \frac{\mathrm{e}^{-x}}{2} - \frac{\mathrm{e}^x}{2} + \frac{\mathrm{e}^{-x}}{2}\right)^3 \\
&= (\mathrm{e}^{-x})^3 = \mathrm{e}^{-3x} \\
&= \frac{\mathrm{e}^{3x}}{2} + \frac{\mathrm{e}^{-3x}}{2} - \frac{\mathrm{e}^{3x}}{2} + \frac{\mathrm{e}^{-3x}}{2} \\
&= \frac{\mathrm{e}^{3x} + \mathrm{e}^{-3x}}{2} - \frac{\mathrm{e}^{3x} - \mathrm{e}^{-3x}}{2} \\
&= \cosh 3x - \sinh 3x
\end{aligned}$$

So that $(\cosh x + \sinh x)^3 = \cosh 3x + \sinh 3x$ and $(\cosh x - \sinh x)^3 = \cosh 3x - \sinh 3x$

Alternatively:

Since $\cosh x + \sinh x = \frac{1}{2}(\mathrm{e}^x + \mathrm{e}^{-x}) + \frac{1}{2}(\mathrm{e}^x - \mathrm{e}^{-x}) = \mathrm{e}^x$

$$(\cosh x + \sinh x)^3 = \mathrm{e}^{3x} = \cosh(3x) + \sinh(3x)$$

Since $\cosh x - \sinh x = \frac{1}{2}(\mathrm{e}^x + \mathrm{e}^{-x}) - \frac{1}{2}(\mathrm{e}^x - \mathrm{e}^{-x}) = \mathrm{e}^{-x}$

$$\begin{aligned}
(\cosh x - \sinh x)^3 &= \mathrm{e}^{-3x} = \cosh(-3x) + \sinh(-3x) \\
&= \cosh 3x - \sinh 3x
\end{aligned}$$

So that $(\cosh x + \sinh x)^3 = \cosh 3x + \sinh 3x$ and $(\cosh x - \sinh x)^3 = \cosh 3x - \sinh 3x$

**15 c)**

$$\begin{aligned}
(\cosh x + \sinh x)^3 + (\cosh x - \sinh x)^3 &= \cosh 3x + \sinh 3x + \cosh 3x - \sinh(3x) \\
&= 2\cosh 3x \\
\therefore 2\cosh 3x &= \cosh^3 x + 3\cosh^2 x \sinh x + 3\cosh x \sinh^2 x + \sinh^3 x
\end{aligned}$$

$$
\begin{aligned}
&\quad + (\cosh^3 x - 3\cosh^2 x \sinh x + 3\cosh x \sinh^2 x - \sinh^3 x) \\
&= 2\cosh^3 x + 6\cosh x \sinh^2 x \\
\therefore \cosh 3x &= \cosh^3 x + 3\cosh x \sinh^2 x \\
(\cosh x + \sinh x)^3 - (\cosh x - \sinh x)^3 &= \cosh 3x + \sinh 3x - \cosh 3x + \sinh 3x \\
&= 2\sinh 3x \\
\therefore 2\sinh 3x &= \cosh^3 x + 3\cosh^2 x \sinh x + 3\cosh x \sinh^2 x + \sinh^3 x \\
&\quad - (\cosh^3 x - 3\cosh^2 x \sinh x + 3\cosh x \sinh^2 x - \sinh^3 x) \\
&= 6\cosh^2 x \sinh x + 2\sinh^3 x \\
\therefore \sinh 3x &= 3\cosh^2 x \sinh x + \sinh^3 x
\end{aligned}
$$

Hence $\cosh 3x = \cosh^3 x + 3\cosh x \sinh^2 x$ and $\sinh 3x = 3\cosh^2 x \sinh x + \sinh^3 x$

**15 d)**

$$\cosh 3x = \cosh^3 x + 3\cosh x \sinh^2 x = 2^3 + 3 \times 2 \times 3 = 26$$

and

$$\sinh 3x = 3\cosh^2 x \sinh x + \sinh^3 x = 3 \times 2^2 \times \sqrt{3} + 3\sqrt{3} = 15\sqrt{3}$$

$\cosh 3x = 26$ and $\sinh 3x = 15\sqrt{3}$

**16 a)**

$$
\begin{aligned}
\mathrm{e}^{\mathrm{i}z} &= \cos z + \mathrm{i}\sin z \\
\cosh(\mathrm{i}z) &= \frac{1}{2}(\mathrm{e}^{\mathrm{i}z} + \mathrm{e}^{-\mathrm{i}z}) \\
&= \frac{1}{2}(\cos z + \mathrm{i}\sin z + \cos(-z) + \mathrm{i}\sin(-z)) \\
&= \frac{1}{2}(\cos z + \mathrm{i}\sin z + \cos z - \mathrm{i}\sin z) \\
&= \frac{1}{2}(2\cos z) = \cos z \quad \text{as req.}
\end{aligned}
$$

**16 b)**

$$\cosh\left(\frac{3\pi}{4}\mathrm{i}\right) = \cos\frac{3\pi}{4} = -\frac{\sqrt{2}}{2}$$

**17 a)**

$$
\begin{aligned}
\mathrm{e}^{\mathrm{i}z} &= \cos z + \mathrm{i}\sin z \\
\sinh(\mathrm{i}z) &= \frac{1}{2}(\mathrm{e}^{\mathrm{i}z} - \mathrm{e}^{-\mathrm{i}z}) \\
&= \frac{1}{2}(\cos z + \mathrm{i}\sin z - \cos(-z) - \mathrm{i}\sin(-z)) \\
&= \frac{1}{2}(\cos z + \mathrm{i}\sin z - \cos z + \mathrm{i}\sin z) \\
&= \frac{1}{2}(2\mathrm{i}\sin z) = \mathrm{i}\sin z \quad \text{as req.}
\end{aligned}
$$

**17 b)**

$$\sinh\left(\frac{5\pi}{6}\mathrm{i}\right) = \mathrm{i}\sin\frac{5\pi}{6} = \frac{1}{2}\mathrm{i}$$

**18 a)**

$$\mathrm{e}^{\mathrm{i}z} = \cos z + \mathrm{i}\sin z$$

$$\begin{aligned}\sinh(\mathrm{i}z) &= \frac{1}{2}(\mathrm{e}^{\mathrm{i}z} - \mathrm{e}^{-\mathrm{i}z}) \\ &= \frac{1}{2}(\cos z + \mathrm{i}\sin z - \cos(-z) - \mathrm{i}\sin(-z)) \\ &= \frac{1}{2}(\cos z + \mathrm{i}\sin z - \cos z + \mathrm{i}\sin z) \\ &= \frac{1}{2}(2\mathrm{i}\sin z) = \mathrm{i}\sin z \quad \text{as req.}\end{aligned}$$

**18 b)**

$\frac{3\pi}{7} = 3\theta$ where $\theta = \frac{\pi}{7}$

Since $(\cos\theta + \mathrm{i}\sin\theta)^3 = \cos(3\theta) + \mathrm{i}\sin(3\theta)$

$$\left(\cos\frac{\pi}{7} + \mathrm{i}\sin\frac{\pi}{7}\right)^3 = \cos\frac{3\pi}{7} + \mathrm{i}\sin\frac{3\pi}{7}$$

Consider the LHS:

$$\begin{aligned}\left(\cos\frac{\pi}{7} + \mathrm{i}\sin\frac{\pi}{7}\right)^3 = \cos^3\left(\frac{\pi}{7}\right) + 3\mathrm{i}\cos^2\left(\frac{\pi}{7}\right)\sin\frac{\pi}{7} \\ + 3\cos\frac{\pi}{7}\sin^2\left(\frac{\pi}{7}\right) - \mathrm{i}\sin^3\left(\frac{\pi}{7}\right)\end{aligned}$$

Compare the imaginary parts:

$$\mathrm{i}\sin\frac{3\pi}{7} = 3\mathrm{i}\cos^2\left(\frac{\pi}{7}\right)\sin\frac{\pi}{7} - \mathrm{i}\sin^3\left(\frac{\pi}{7}\right)$$

But since $\sinh(\mathrm{i}z) = \mathrm{i}\sin z$ then $\sinh\left(\frac{3\pi}{7}\mathrm{i}\right) = 3\mathrm{i}\cos^2\left(\frac{\pi}{7}\right)\sin\frac{\pi}{7} - \mathrm{i}\sin^3\left(\frac{\pi}{7}\right)$

**19)**

Since $\cos 3\theta + \mathrm{i}\sin 3\theta = (\cos\theta + \mathrm{i}\sin\theta)^3$ Equate real parts:

$$\begin{aligned}\cos 3\theta &= \cos^3\theta - 3\cos\theta\sin^2\theta \\ &= \cos^3\theta - 3\cos\theta(1 - \cos^2\theta) \\ &= \cos^3\theta - 3\cos\theta + 3\cos^3\theta \\ &= 4\cos^3\theta - 3\cos\theta\end{aligned}$$

Considering $x = \cos\frac{4\pi}{9}$ then let $\theta = \frac{4\pi}{9}$ so that $3\theta = \frac{4\pi}{3}$ So $\cos 3\theta = \cos\frac{4\pi}{3} = -\frac{1}{2}$

So using $x = \cos\theta$ in $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$:

$$\begin{aligned}\cos\frac{4\pi}{3} &= 4x^3 - 3x \\ -\frac{1}{2} &= 4x^3 - 3x \\ \therefore 8x^3 - 6x + 1 &= 0 \quad \text{as req.}\end{aligned}$$

**20 a)**

$$(\cos\theta + \mathrm{i}\sin\theta)^5 = \cos^5\theta + 5\mathrm{i}\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta - 10\mathrm{i}\cos^2\theta\sin^3\theta + 5\cos\theta\sin^4\theta + \mathrm{i}\sin^5\theta$$

Since $\cos 5\theta + \mathrm{i}\sin 5\theta = (\cos\theta + \mathrm{i}\sin\theta)^5$ Equate real parts:

$$\begin{aligned}
\cos 5\theta &= \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta \\
&= \cos^5\theta - 10\cos^3\theta(1-\cos^2\theta) + 5\cos\theta(1-\cos^2\theta)(1-\cos^2\theta) \\
&= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta(1 - 2\cos^2\theta + \cos^4\theta) \\
&= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta - 10\cos^3\theta + 5\cos^5\theta \\
&= 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta \quad \text{as req.}
\end{aligned}$$

**20 b)**

$$16\cos^5\theta - 20\cos^3\theta + 5\cos\theta = 0$$

Let $x = \cos\theta$ so that $16x^5 - 20x^3 + 5x = 0$

$$16x^5 - 20x^3 + 5x = x(16x^4 - 20x^2 + 5)$$

So that

$$x = 0 \text{ or } x^2 = \frac{20 \pm \sqrt{400 - 320}}{32} = \frac{20 \pm \sqrt{80}}{32} = \frac{20 \pm 4\sqrt{5}}{32} = \frac{5 \pm \sqrt{5}}{8}$$

So that $\cos\theta = 0$ or $\cos^2\theta = \frac{5+\sqrt{5}}{8}$ or $\cos^2\theta = \frac{5-\sqrt{5}}{8}$

Solving $\cos 5\theta = 0$ for $0 \le \theta \le \pi$ gives $\frac{\pi}{10}$, $\frac{3\pi}{10}$, $\frac{5\pi}{10}$, $\frac{7\pi}{10}$ or $\frac{9\pi}{10}$

Since $\frac{\pi}{10}$ and $\frac{3\pi}{10}$ are both in the first quadrant, then for cosine, $\frac{\pi}{10}$ must give the larger solution and $\frac{3\pi}{10}$ must give the smaller solution.

So that

$$\cos^2\left(\frac{\pi}{10}\right) = \frac{5+\sqrt{5}}{8} \text{ and } \cos^2\left(\frac{3\pi}{10}\right) = \frac{5-\sqrt{5}}{8} \quad \text{as req.}$$

**21 a)**

$$\begin{aligned}
&(\cos\theta + \mathrm{i}\sin\theta)^6 \\
&= \cos^6\theta + 6\mathrm{i}\cos^5\theta\sin\theta - 15\cos^4\theta\sin^2\theta - 20\mathrm{i}\cos^3\theta\sin^3\theta + 15\cos^2\theta\sin^4\theta + 6\mathrm{i}\cos\theta\sin^5\theta - \sin^6\theta
\end{aligned}$$

Since $\cos 6\theta + \mathrm{i}\sin 6\theta = (\cos\theta + \mathrm{i}\sin\theta)^6$ Equate real parts:

$$\begin{aligned}
&\cos 6\theta \\
&= \cos^6\theta - 15\cos^4\theta\sin^2\theta + 15\cos^2\theta\sin^4\theta - \sin^6\theta \\
&= \cos^6\theta - 15\cos^4\theta(1-\cos^2\theta) + 15\cos^2\theta(1-\cos^2\theta)^2 - (1-\cos^2\theta)^3 \\
&= \cos^6\theta - 15\cos^4\theta + 15\cos^6\theta + 15\cos^2\theta(1 - 2\cos^2\theta + \cos^4\theta) - (1 - 3\cos^2\theta + 3\cos^4\theta - \cos^6\theta) \\
&= \cos^6\theta - 15\cos^4\theta + 15\cos^6\theta + 15\cos^2\theta - 30\cos^4\theta + 15\cos^6\theta - 1 + 3\cos^2\theta - 3\cos^4\theta + \cos^6\theta \\
&= 32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1 \quad \text{as req.}
\end{aligned}$$

**21 b)**

$$32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1 = 0$$

Let $x = \cos^2\theta$ so that $32x^3 - 48x^2 + 18x - 1 = 0$

Let $f(x) = 32x^3 - 48x^2 + 18x - 1$

$$f\left(\frac{1}{2}\right) = 4 - 12 + 9 - 1 = 0 \quad \therefore (2x-1) \text{ is a factor}$$

$$32x^3 - 48x^2 + 18x - 1 = (2x-1)(16x^2 - 16x + 1)$$

So that

$$x = \frac{1}{2} \text{ or } x = \frac{16 \pm \sqrt{256-64}}{32} = \frac{16 \pm \sqrt{192}}{32} = \frac{16 \pm 8\sqrt{3}}{32} = \frac{2 \pm \sqrt{3}}{4}$$

So that $\cos^2\theta = \frac{1}{2}$ or $\cos^2\theta = \frac{2+\sqrt{3}}{4}$ or $\cos^2\theta = \frac{2-\sqrt{3}}{4}$

Solving $\cos(6\theta) = 0$ for $0 \le \theta \le \pi$ gives $\frac{\pi}{12}$, $\frac{3\pi}{12}$, $\frac{5\pi}{12}$, $\frac{7\pi}{12}$ or $\frac{9\pi}{12}$

Since $\frac{\pi}{12}$ and $\frac{5\pi}{12}$ are both in the first quadrant, then for cosine, $\frac{\pi}{12}$ must give the larger solution and $\frac{5\pi}{12}$ must give the smaller solution.

So that

$$\cos^2\left(\frac{\pi}{12}\right) = \frac{2+\sqrt{3}}{4} \text{ and } \cos^2\left(\frac{5\pi}{12}\right) = \frac{2-\sqrt{3}}{4} \quad \text{as req.}$$

**22 a)**

$$(\cos\theta + \mathrm{i}\sin\theta)^5 = \cos^5\theta + 5\mathrm{i}\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta - 10\mathrm{i}\cos^2\theta\sin^3\theta + 5\cos\theta\sin^4\theta + \mathrm{i}\sin^5\theta$$

Since $\cos 5\theta + \mathrm{i}\sin 5\theta = (\cos\theta + \mathrm{i}\sin\theta)^5$ Equate real parts:

$$\begin{aligned}
\cos 5\theta &= \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta \\
&= \cos^5\theta - 10\cos^3\theta(1-\cos^2\theta) + 5\cos\theta(1-\cos^2\theta)(1-\cos^2\theta) \\
&= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta(1 - 2\cos^2\theta + \cos^4\theta) \\
&= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta - 10\cos^3\theta + 5\cos^5\theta \\
&= 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta
\end{aligned}$$

$$\therefore \cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$$

**22 b)**

Solving $\cos 5\theta = 1$:

$$16\cos^5\theta - 20\cos^3\theta + 5\cos\theta = 1$$

Let $x = \cos\theta$ so that $16x^5 - 20x^3 + 5x - 1 = 0$

$$\begin{aligned}
16x^5 - 20x^3 + 5x - 1 &= (x-1)(16x^4 + 16x^3 - 4x^2 - 4x + 1) \\
&= (x-1)(4x^2 + 2x - 1)^2
\end{aligned}$$

$$\therefore (x-1)(4x^2 + 2x - 1)^2 = 0 \quad \text{as req.}$$

**22 c)**

So that

$$x = 1 \text{ or } x = \frac{-2 \pm \sqrt{4+16}}{8} = \frac{-2 \pm 2\sqrt{5}}{8} = \frac{-1 \pm \sqrt{5}}{4}$$

So that $\cos\theta = 1$ or $\cos\theta = \frac{-1+\sqrt{5}}{4}$ or $\cos\theta = \frac{-1-\sqrt{5}}{4}$

Solving $\cos 5\theta = 1$ for $0 \leq \theta \leq 180°$ gives 0, 72°, or 144°

Since 72° is in the first quadrant and 144° is in the second quadrant, 72° must give the positive solution and 144° must give the negative solution.

$$\therefore \cos 72° = \frac{-1+\sqrt{5}}{4} \text{ and } \cos 144° = \frac{-1-\sqrt{5}}{4} \quad \text{as req.}$$

**23 a)**

Since $\cos 2A = 1 - 2\sin^2 A$ then $\cos\theta = 1 - 2\sin^2\left(\frac{\theta}{2}\right)$

Replace $\theta$ with $n\theta$:

$$\cos n\theta = 1 - 2\sin^2\left(\frac{n\theta}{2}\right)$$

**23 b)**

Since $\sin 2A = 2\sin A\cos A$ then $\sin\theta = 2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)$

Replace $\theta$ with $n\theta$:

$$\sin n\theta = 2\sin\left(\frac{n\theta}{2}\right)\cos\left(\frac{n\theta}{2}\right)$$

**23 c)**

$$\sum_{r=1}^{n} \cos(r\theta) = \text{Re}[(\cos\theta + \text{i}\sin\theta) + (\cos\theta + \text{i}\sin\theta)^2 + \cdots + (\cos\theta + \text{i}\sin\theta)^n]$$

This is a geometric series with $a = \cos\theta + \text{i}\sin\theta$ and $r = \cos\theta + \text{i}\sin\theta$

So use $S_n = \frac{a(1-r^n)}{1-r}$

$$\frac{(\cos\theta + \text{i}\sin\theta)(1 - (\cos\theta + \text{i}\sin\theta)^n)}{1 - (\cos\theta + \text{i}\sin\theta)} = \frac{(\cos\theta + \text{i}\sin\theta)(1 - \cos n\theta - \text{i}\sin n\theta)}{1 - \cos\theta - \text{i}\sin\theta}$$

Rationalise the denominator:

$$\frac{(\cos\theta + \text{i}\sin\theta)(1 - \cos n\theta - \text{i}\sin n\theta)}{1 - \cos\theta - \text{i}\sin\theta} \times \frac{1 - \cos\theta + \text{i}\sin\theta}{1 - \cos\theta + \text{i}\sin\theta}$$

Which gives a denominator of

$$\begin{aligned}(1 - \cos\theta - \text{i}\sin\theta)(1 - \cos\theta + \text{i}\sin\theta) &= (1 - \cos\theta)^2 + \sin^2\theta \\ &= 1 - 2\cos\theta + \cos^2\theta + \sin^2\theta = 2 - 2\cos\theta\end{aligned}$$

So this gives:

$$\frac{(\cos\theta + \text{i}\sin\theta)(1 - \cos n\theta - \text{i}\sin n\theta)(1 - \cos\theta + \text{i}\sin\theta)}{2 - 2\cos\theta}$$

Expand brackets:

$$\begin{aligned}(\cos\theta + \mathrm{i}\sin\theta)(1 - \cos\theta + \mathrm{i}\sin\theta) &= \cos\theta - \cos^2\theta + \mathrm{i}\cos\theta\sin\theta + \mathrm{i}\sin\theta - \mathrm{i}\cos\theta\sin\theta - \sin^2\theta \\ &= \cos\theta + \mathrm{i}\sin\theta - 1\end{aligned}$$

So that the fraction becomes:

$$\frac{(\cos\theta + \mathrm{i}\sin\theta - 1)(1 - \cos n\theta - \mathrm{i}\sin n\theta)}{2 - 2\cos\theta}$$

Expand brackets:

$$\begin{aligned}(\cos\theta + \mathrm{i}\sin\theta - 1)(1 - \cos n\theta - \mathrm{i}\sin n\theta) = \cos\theta - \cos\theta\cos n\theta - \mathrm{i}\cos\theta\sin n\theta + \mathrm{i}\sin\theta \\ - \mathrm{i}\sin\theta\cos n\theta + \sin\theta\sin n\theta - 1 + \cos n\theta + \mathrm{i}\sin n\theta\end{aligned}$$

Considering only the real parts:

$$\begin{aligned}\sum_{r=1}^{n}\cos(r\theta) &= \frac{\cos\theta - \cos\theta\cos n\theta + \sin\theta\sin n\theta - 1 + \cos n\theta}{2 - 2\cos\theta} \\ &= \frac{\cos\theta - 1 - \cos\theta\cos n\theta + \sin\theta\sin n\theta + \cos n\theta}{2 - 2\cos\theta} \\ &= \frac{(\cos\theta - 1)(1 - \cos n\theta) + \sin\theta\sin n\theta}{2 - 2\cos\theta}\end{aligned}$$

We know $\cos\theta = 1 - 2\sin^2\left(\frac{\theta}{2}\right)$ and $\sin\theta = 2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)$,

and also $\cos n\theta = 1 - 2\sin^2\left(\frac{n\theta}{2}\right)$ and $\sin n\theta = 2\sin\left(\frac{n\theta}{2}\right)\cos\left(\frac{n\theta}{2}\right)$

So that:

$$\begin{aligned}\cos\theta - 1 &= -2\sin^2\left(\frac{\theta}{2}\right) \\ 1 - \cos n\theta &= 2\sin^2\left(\frac{n\theta}{2}\right) \\ 2 - 2\cos\theta &= 4\sin^2\left(\frac{\theta}{2}\right) \\ \sin\theta\sin n\theta &= 4\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)\sin\left(\frac{n\theta}{2}\right)\cos\left(\frac{n\theta}{2}\right)\end{aligned}$$

Consider the numerator only:

$$\begin{aligned}&(\cos\theta - 1)(1 - \cos n\theta) + \sin\theta\sin n\theta \\ &= -4\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{n\theta}{2}\right) \\ &\quad + 4\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)\sin\left(\frac{n\theta}{2}\right)\cos\left(\frac{n\theta}{2}\right) \\ &= 4\sin\left(\frac{\theta}{2}\right)\sin\left(\frac{n\theta}{2}\right)\left[-\sin\left(\frac{\theta}{2}\right)\sin\left(\frac{n\theta}{2}\right) + \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{n\theta}{2}\right)\right]\end{aligned}$$

Using the compound angle formula, $\cos(A + B) = \cos A\cos B - \sin A\sin B$

$$= 4\sin\left(\frac{\theta}{2}\right)\sin\left(\frac{n\theta}{2}\right)\cos\left(\frac{\theta}{2} + \frac{n\theta}{2}\right)$$

Therefore

$$\sum_{r=1}^{n}\cos(r\theta) = \frac{4\sin\left(\frac{\theta}{2}\right)\sin\left(\frac{n\theta}{2}\right)\cos\left(\frac{\theta}{2} + \frac{n\theta}{2}\right)}{4\sin^2\left(\frac{\theta}{2}\right)}$$

$$= \frac{\sin\left(\frac{n\theta}{2}\right)\cos\left(\frac{(n+1)\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}$$

As req.

**23 d)**

Similarly:

Considering only the imaginary parts:

$$\sum_{r=1}^{n}\sin(r\theta) = \frac{-\cos\theta\sin n\theta + \sin\theta - \sin\theta\cos n\theta + \sin n\theta}{2-2\cos\theta}$$
$$= \frac{(1-\cos\theta)\sin n\theta + \sin\theta(1-\cos n\theta)}{2-2\cos\theta}$$

Since

$$1-\cos\theta = 2\sin^2\left(\frac{\theta}{2}\right)$$
$$1-\cos n\theta = 2\sin^2\left(\frac{n\theta}{2}\right)$$
$$2-2\cos\theta = 4\sin^2\left(\frac{\theta}{2}\right)$$

This gives

$$\frac{2\sin^2\left(\frac{\theta}{2}\right)\sin n\theta + 2\sin\theta\sin^2\left(\frac{n\theta}{2}\right)}{4\sin^2\left(\frac{\theta}{2}\right)}$$
$$= \frac{\sin^2\left(\frac{\theta}{2}\right)\sin n\theta + \sin\theta\sin^2\left(\frac{n\theta}{2}\right)}{2\sin^2\left(\frac{\theta}{2}\right)}$$

Since $\sin\theta = 2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)$ and $\sin n\theta = 2\sin\left(\frac{n\theta}{2}\right)\cos\left(\frac{n\theta}{2}\right)$

$$= \frac{2\sin^2\left(\frac{\theta}{2}\right)\sin\left(\frac{n\theta}{2}\right)\cos\left(\frac{n\theta}{2}\right) + 2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)\sin^2\left(\frac{n\theta}{2}\right)}{2\sin^2\left(\frac{\theta}{2}\right)}$$
$$= \frac{\sin\left(\frac{\theta}{2}\right)\sin\left(\frac{n\theta}{2}\right)\cos\left(\frac{n\theta}{2}\right) + \cos\left(\frac{\theta}{2}\right)\sin^2\left(\frac{n\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}$$
$$= \frac{\sin\left(\frac{n\theta}{2}\right)\left[\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{n\theta}{2}\right) + \cos\left(\frac{\theta}{2}\right)\sin\left(\frac{n\theta}{2}\right)\right]}{\sin\left(\frac{\theta}{2}\right)}$$

Using the compound angle formula, $\sin(A+B) = \sin A\cos B + \cos A\sin B$

$$\sum_{r=1}^{n}\sin(r\theta) = \frac{\sin\left(\frac{n\theta}{2}\right)\sin\left(\frac{\theta}{2}+\frac{n\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}$$
$$= \frac{\sin\left(\frac{n\theta}{2}\right)\sin\left(\frac{(n+1)\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}$$

As req.